

STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 5- Binomial Random Variable & Distribution

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Session Outline

- 1 Binomial Random Variable
 - Binomial Experiment
- 2 Binomial Distribution
- 3 Moments
- 4 Computation of Probabilities
- 5 Cumulative Distribution Function

Session Overview

In the preview session students were introduced to Bernoulli random variable and its probability distribution, which serve the bases for the binomial random variable. This session introduces students to Binomial probability distribution. The experiment that gives rise to the Binomial random variable and properties of the distribution will be discussed. Computation of the probabilities using the Binomial probability mass function and theoretical derivation of the mean and variance of the Binomial random variable will be presented.

Learning Outcomes

At the end of the session, students will be able to:

- 1 Explain and describe the experiment leading to Binomial random variable.
- 2 State the probability mass function of the Binomial distribution and show that it is a probability function.
- 3 Prove the expectation and variance of Binomial distribution.
- 4 Compute probabilities and moments using the Binomial distribution function.

Binomial random variable

- Situations under which the Binomial distribution arises:
 - 1 There are two possible outcomes in each trial, ie. *Success* and *Failure*.
 - 2 There are fixed number of trials or repetitions, ie. n .
 - 3 Trials or repetitions are independent of each another.
 - 4 The probabilities of *Success* and *Failure* are constant throughout the trials or repetitions of the experiment.

Binomial Experiment

- Let X be the number of successes in n fixed trials of an experiment.
- Let $P(\text{Success}) = p$; $0 < p < 1$ and $P(\text{Failure}) = 1 - p = q$.
- Suppose the event of a success is A , then the event of a failure is $\bar{A}$
- Suppose further that there are x trials resulting in successes, then there will be $n - x$ failures.
- Suppose a typical outcome of the experiment is as follows:

Trials	1	2	3	...	x^{th}	$(x + 1)^{th}$	...	n^{th}
Outcome	A	A	A	...	A	$\bar{A}$	...	$\bar{A}$

- Now, there are several arrangements of the outcome for which x successes and $n - x$ failures results.

Binomial Distribution

- The probability for x successes is given as:

$$P(X = x) = \left(\begin{array}{c} \text{No. of arrangements for } x \\ \text{successes \& } (n - x) \text{ failures} \end{array} \right)$$

$$\times P(x \text{ Successes \& } (n - x) \text{ Failures})$$

- Since the trials are independent, we have the probability of x successes as:

$$P(X = x) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

Binomial Probability Mass Function

$$P(X = x) = \binom{n}{x} p^x q^{n-x},$$

where $x = 0, 1, 2, \dots, n$, $0 < x < 1$ and $q = 1 - p$.

Binomial Distribution

- The Binomial probability mass function is known as the Binomial distribution with parameters n and p .
- Notation: If the random variable X has a Binomial distribution with parameters n and p , it is denoted as $X \sim b(n, p)$.
- If $X \sim b(n, p)$, then

$$P(X = x) = b(x; n, p) = \binom{n}{x} p^x q^{n-x}.$$

- Note that, the Binomial is an approximate value for a random variable which measures the number of occurrences of a given attributes in a sample of n individuals selected at random from a given population.

Legitimacy Binomial Distribution

Theorem

If the random variable $X \sim b(n, p)$, then its distribution function is **legitimate**.

Proof

(i). The probability mass function is greater than or equal to zero for all values of x , i.e. $P(X = x) \geq 0, \forall x$.

$$P(X = x) = \binom{n}{x} p^x q^{n-x},$$

$$\Rightarrow \binom{n}{x} = \frac{n!}{x!(n-x)!} > 0, \forall x$$

Also, $p^x > 0, \forall x$, since $0 < p < 1$ and $q^{n-x} > 0, \forall x$.

$$\therefore P(X = x) > 0, \forall x.$$

Proof Cont'd

$$(ii.) \sum_{all\ x} P(X = x) = 1$$

i.e.

$$\sum_{x=0}^n P(X = x) = \sum_{x=0}^n \binom{n}{x} p^x q^{n-x},$$

Using the **Binomial Theorem**, which state that,

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^r b^{n-r},$$

for a and b are real numbers or constants and $n \in \mathbb{Z}^+$, we have,

$$\sum_{x=0}^n P(X = x) = (p + q)^n = 1^n = 1 \quad \square$$

Hence the Proof.

Characteristics of Binomial Distribution

Theorem

If the random variable $X \sim b(n, p)$, then

$$E(X) = np \quad \text{and} \quad \text{Var}(X) = npq.$$

Proof of Expectation

By definition:

$$E(X) = \sum_{x=0}^n X P(X = x) = \sum_{x=0}^n X \binom{n}{x} p^x q^{n-x}$$

$$E(X) = \sum_{x=0}^n X \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

$$= \sum_{x=0}^n X \frac{n!}{x(x-1)!(n-x)!} p^x q^{n-x}$$

Expectation Proof Cont'd

$$E(X) = \sum_{x=1}^n \frac{n!}{(x-1)!(n-x)!} p^x q^{n-x}$$

i.e. dividing by X and then the summation now starts from 1, for the expression to be defined.

$$\begin{aligned} E(X) &= \sum_{x=1}^n \frac{n(n-1)!}{(x-1)!(n-x)!} p^x q^{n-x} \\ &= \sum_{x=1}^n n \binom{n-1}{x-1} p^x q^{n-x} \end{aligned}$$

Expectation Proof Cont'd

Note: by changing variable, on letting $x = y + 1$, we have

$$E(X) = n \sum_{y=0}^{n-1} \binom{n-1}{y} p^{y+1} q^{n-y-1}$$

$$E(X) = np \sum_{y=0}^{n-1} \binom{n-1}{y} p^y q^{(n-1)-y}$$

$$E(X) = np(p + q)^{n-1} = np,$$

by the Binomial Theorem.



Proof of Variance

By definition: $\text{Var}(X) = E(X^2) - [E(X)]^2$

$$\Rightarrow E(X^2) = \sum_{x=0}^n x^2 \binom{n}{x} p^x q^{n-x}$$

Note that: $X^2 = X(X-1) + X$

$$\begin{aligned}\Rightarrow E(X^2) &= \sum_{x=0}^n [x(x-1) + x] \frac{n!}{x!(n-x)!} p^x q^{n-x} \\ &= \sum_{x=0}^n x(x-1) \frac{n!}{x(x-1)(x-2)!(n-x)!} p^x q^{n-x} \\ &\quad + \sum_{x=0}^n x \frac{n!}{x!(n-x)!} p^x q^{n-x}\end{aligned}$$

Proof of Variance Cont'd

$$\begin{aligned} E(X^2) &= \sum_{x=2}^n \frac{n(n-1)(n-2)!}{(x-2)!(n-x)!} p^x q^{n-x} + E(X) \\ &= \sum_{x=2}^n n(n-1) \binom{n-2}{x-2} p^x q^{n-x} + E(X) \end{aligned}$$

Note: by changing variable, on letting $x = y + 2$, we have

$$E(X^2) = n(n-1) \sum_{y=0}^{n-2} \binom{n-2}{y} p^{y+2} q^{n-y-2} + E(X)$$

$$E(X^2) = n(n-1)p^2 \sum_{y=0}^{n-2} \binom{n-2}{y} p^y q^{(n-2)-y} + E(X)$$

Proof of Variance Cont'd

$$E(X^2) = n(n-1)p^2(p+q)^{n-2} + E(X)$$

$$\therefore E(X^2) = n(n-1)p^2 + np$$

Hence the Variance of the random variable X is:

$$\text{Var}(X) = n(n-1)p^2 + np - (np)^2$$

$$= np\{(n-1)p + 1 - np\} = np(1-p)$$

$$\therefore \text{Var}(X) = npq$$



Applications

- Example 1: Suppose the probability of defective items in the production process of car batteries is 0.2. If ten items are selected from the process, find
 - a. the expected number of good items.
 - b. probability that none of the items is none defective.
 - c. probability that at least three are defective.

Solution

- a. Let the random variable X be the number of good items or batteries, the $X \sim b(n = 10, p = 0.8)$.

Therefore, the expected number of good batteries produced is:

$$E(X) = np = 10 \times 0.8 = 8.$$



Example 1 Cont'd

Solution Cont'd

- ⓑ Let the random variable Y be the number of defective items or batteries, the $Y \sim b(n = 10, p = 0.2)$.

Therefore, the required probability distribution function is

$$P(Y = y) = \binom{10}{y} (0.2)^y \times (0.8)^{10-y}$$

$$\begin{aligned}\therefore P(Y = 0) &= \binom{10}{0} (0.2)^0 \times (0.8)^{10-0} \\ &= (0.8)^{10} = 0.107, \quad 3d.p.\end{aligned}$$



Theorem

- Before we answer (c), let look at this Theorem.
- **Theorem:** If $X \sim b(n, p)$, then $Y = (n - X) \sim b(n, q)$, where $q = 1 - p$, and
 - i $P(X = x) = P(Y = n - x)$
 - ii $P(X \geq x) = P(Y \leq n - x)$.

Solution (c)

The required probability of at least three are defective is:

$$\begin{aligned}P(Y \geq 3) &= 1 - P(Y < 3) = 1 - P(Y \leq 2) \\&= 1 - \sum_{y=0}^2 \binom{10}{y} (0.2)^y \times (0.8)^{10-y} \\&= 1 - [P(Y = 0) + P(Y = 1) + P(Y = 2)] \\&= 1 - 0.89254 = 0.10746\end{aligned}$$

Example 2

Question: Suppose a ball is drawn at random from a box containing three white and three black balls. After a ball is drawn, it is then replaced and another drawn. What is the probability that of the first four balls drawn, exactly two are white?

Solution

Let X be the number of white balls drawn, then $X \sim b(4, p = 1/2)$, ie. $p = \left(\frac{3}{6}\right)$ and $n = 4$, the number of balls drawn.

$$\therefore P(X = x) = \binom{4}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}$$

$$\begin{aligned} P(X = 2) &= \binom{4}{x} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{4-2} \\ &= 0.375 \end{aligned}$$

- 1 On a multiple choice test with four possible answers for each of the fifteen questions. What is the probability that a student would score four or more correct answers just by guessing?
Assume there is only one correct answer for each questions and the questions are independent.

Cumulative Distribution Function

- The cumulative distribution function of the Binomial distribution is denoted as $B(x; n, p)$ and defined by

$$\begin{aligned} P(X \leq x) &= \sum_{r=0}^x P(X = r) = \sum_{r=0}^x b(r; n, p) \\ &= \sum_{r=0}^x \binom{n}{r} p^r q^{n-r}, \quad r = 0, 1, 2, \dots, n \end{aligned}$$

Example 3

Suppose that the random variable $X \sim b(10, 0.3)$, calculate the following probabilities using the Binomial probability tables:

- i $P(X \leq 7)$
- ii $P(X > 4)$
- iii $P(X = 5)$.

Solution

Using the Binomial probability tables, we have the following:

- i $P(X \leq 7) = B(7; 10, 0.3) = 0.9984$
- ii $P(X > 4) = 1 - P(X \leq 4) = 1 - B(4; 10, 0.3)$
 $= 1 - 0.8497 = 0.1503.$
- iii $P(X = 5) = P(X \leq 5) - P(X \leq 4)$
 $= 0.9527 - 0.8497 = 0.103.$

Trial Questions

- 1 Suppose you play a game that you can only either win or lose. The probability that you win any game is 55%, and the probability that you lose is 45%. Each game you play is independent. If you play the game 20 times, write the function that describes the probability that you win 15 of the 20 times and calculate it.
- 2 A trainer is teaching a dolphin to do tricks. The probability that the dolphin successfully performs the trick is 35%, and the probability that the dolphin does not successfully perform the trick is 65%. Out of 20 attempts, you want to find the probability that the dolphin succeeds 12 times. Compute the probability.

Trial Questions:

- 3
 - i A fair coin is flipped 15 times. Each flip is independent. What is the probability of getting more than ten heads?
 - ii A fair, six-sided die is rolled ten times. Each roll is independent. Find the probability of rolling a one more than three times.
 - iii Suppose that about 32% of students participate in a community volunteer program outside of school. If 30 students are selected at random, find the probability that at most 14 of them participate in a community volunteer program outside of school.

Trial Questions:

- 4 According to a Gallup poll, 60% of American adults prefer saving over spending. Let X denote the number of American adults out of a random sample of 50 who prefer saving to spending.
- i What is the probability distribution for X ?
 - ii Find the following probabilities:
 - a. the probability that 25 adults in the sample prefer saving over spending.
 - b. the probability that at most 20 adults prefer saving.
 - c. the probability that more than 30 adults prefer saving.
 - iii Calculate the mean and standard deviation of X .

THANK YOU!!!